MathEXplained Magazine

April Crossword Solutions

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1: **1**	2: **7**
3: **2**	**8**

ONE ACROSS

If N is the smallest positive integer satisfying these conditions, find the sum of the digits of N. We solve the system

$$N \equiv 4 \pmod 5, \qquad N \equiv 5 \pmod 7, \qquad N \equiv 8 \pmod 9.$$

By the Chinese Remainder Theorem, since $5, 7, 9$ are pairwise coprime, a unique solution exists modulo $5 \cdot 7 \cdot 9 = 315$. Write $N = 5a + 4$. Substituting into the second congruence:

$$5a + 4 \equiv 5 \pmod 7 \quad \Rightarrow \quad 5a \equiv 1 \pmod 7.$$

Since $5 \cdot 3 = 15 \equiv 1 \pmod 7$, we have $5^{-1} \equiv 3 \pmod 7$, so $a \equiv 3 \pmod 7$. Thus $a = 7b + 3$ and

$$N = 5(7b + 3) + 4 = 35b + 19.$$

Substitute into $N \equiv 8 \pmod 9$:

$$35b + 19 \equiv 8 \pmod 9.$$

Since $35 \equiv -1 \pmod 9$ and $19 \equiv 1 \pmod 9$:

$$-b + 1 \equiv 8 \pmod 9 \quad \Rightarrow \quad b \equiv -7 \equiv 2 \pmod 9.$$

Take $b = 2$:

$$N = 35(2) + 19 = 89.$$

$$\begin{aligned} 89 &= 5 \cdot 17 + 4, \\ 89 &= 7 \cdot 12 + 5, \\ 89 &= 9 \cdot 9 + 8. \end{aligned}$$

All three congruences hold. Thus, the answer is $8 + 9 = \boxed{17}$.

ONE DOWN

Let the base side length be $s = 5$ and the height of the lantern be h. The total surface area of a right rectangular prism with a square base is the sum of the areas of the two square bases and the four identical rectangular sides. We can express this as:

$$\text{Surface Area} = 2s^2 + 4sh$$

Substituting the known values for the surface area and the side length into our equation gives:

$$290 = 2(5^2) + 4(5)h$$

$$290 = 2(25) + 20h$$

$$290 = 50 + 20h$$

Subtracting 50 from both sides of the equation leaves:

$$240 = 20h$$

Dividing by 20 to solve for the height:

$$h = 12$$

The height of the stained-glass lantern is

$$\boxed{12}.$$

TWO DOWN

To travel from $(0,0)$ to $(11,2)$, the starship must make a total of 11 jumps East and 2 jumps North in some order. The total number of jumps required for any route is:

$$11 + 2 = 13 \text{ jumps}$$

Any valid route can be represented as a sequence of 13 jumps, where exactly 2 of those jumps must be directed North. The remaining 11 jumps will automatically be East. To find the total number of unique routes, we just need to find the number of ways to choose the 2 North jumps out of the 13 total jumps. This is given by the binomial coefficient $\binom{13}{2}$. We calculate this combination as:

$$\binom{13}{2} = \frac{13!}{2!(13-2)!} = \frac{13 \cdot 12}{2 \cdot 1}$$
$$13 \cdot 6 = 78$$

The navigator can plot exactly

$$\boxed{78}$$

unique routes.

THREE ACROSS

Let the mean of S be

$$\mu = \frac{1}{n} \sum_{k \in S} k.$$

We are given

$$\sum_{k \in S} (k-8)^2 = 122n$$

and

$$\sum_{k \in S} (k+6)^2 = 486n.$$

Subtract the first equation from the second:

$$\sum_{k \in S} \left((k+6)^2 - (k-8)^2\right) = 486n - 122n.$$

Now simplify the expression inside the sum:

$$(k+6)^2 - (k-8)^2 = (k^2 + 12k + 36) - (k^2 - 16k + 64) = 28k - 28.$$

Therefore,

$$\sum_{k \in S} (28k - 28) = 364n \Rightarrow 28 \sum_{k \in S} (k-1) = 364n \Rightarrow \sum_{k \in S} (k-1) = 13n.$$

Thus,

$$\sum_{k \in S} k - n = 13n \Rightarrow \sum_{k \in S} k = 14n.$$

Therefore, the mean of S is

$$\mu = \frac{1}{n} \sum_{k \in S} k = \frac{14n}{n} = 14.$$

Twice the mean is $2 \cdot 14 = \boxed{28}$.